Head Motion Tracker

Background

The Behavioral and Neurodata Core needs adolescent participants, prone to hyperactive movement, to practice staying still before MRI scans. To reduce lab costs, rather than interface with an $8,000 machine that uses a fixed transmitter, our device uses multiple accelerometers to detect motion. If acceleration is greater than an adjustable threshold, a haptic motor buzzes to inform the user.

Requirements & Assumptions

The Head Motion Tracker sends data from multiple SeeedSense MCUs to the Motion Trainer Electron.js app over BLE. Based on configuration in the Motion Trainer app, this data is fused together. Then, BLE is used to tell the Arduino whether to drive the Haptic Motor. The motor is driven when users exceed a certain threshold of acceleration. Additionally, an error light detects if a device is disconnected from the Motion Trainer. We assume that the haptic motor is noticeable for users. We also assume that users will understand that vibration is something to avoid in their usage.